

Problem: From the following table calculate UN-Age-Sex accuracy index and comment.

	Age Group	Population	
		Male	Female
	0-4	1022001	965554
	5-9	964951	916269
	10-14	841146	792129
	15-19	531428	503091
	20-24	318826	470133
	25-29	445106	429185
	30-34	423287	377645
	35-39	400610	325684
	40-44	336099	265734
	45-49	270331	209086
	50-54	236986	186234
	55-59	163501	143373
	60-64	108910	105412
	65-69	67351	74716
	70-74	36557	48300

Solution: UN-Age-Sex accuracy index has three components;

(a) Average sex ratio score (S):

This score is obtained by first calculating the sex ratio at each group. Then successive differences irrespective of sign are added and averaged.

$$\text{Age-specific sex ratio} = \frac{{}_5P_x^m}{{}_5P_x^f} \times 100$$

where, ${}_5P_x^m$ = males aged x to $x+5$ and

${}_5P_x^f$ = females aged x to $x+5$

(b) Average male age ratio score (M):

For each age group for males, calculate the age ratios computed as

$$\text{Age ratio} = \frac{{}_5P_x}{\frac{1}{2}({}_5P_{x-5} + {}_5P_{x+5})} \times 100$$

The deviations from unity irrespective of sign are added and averaged (M).

(c) Average female age ratio score (F):

For each age group for females, the age ratios are calculated using the same formulae as for males.

$$\text{Age-Ratio} = \frac{{}_5P_x}{\frac{1}{2}({}_5P_{x-5} + {}_5P_{x+5})} \times 100$$

The deviations from unity irrespective of sign are added and averaged (F).

The index is then computed as :

$$\text{UNAI} = 3(S) + M + F$$

The reported age-sex data for a given population is presumed to be:

1. "accurate" if the age sex accuracy index is between 0 and 19.9,
2. "inaccurate" if the index is between 20 and 39.9, and
3. "highly inaccurate" if the index is above 40.

The relevant calculation is given in the following table by using MS Excel.

Age Group	Population		Sex Ratio	Successive Difference
	Male	Female		
0-4	1022001	965557	105.9461	
5-9	964956	916239	105.3136	0.5324
10-14	841146	799179	105.2466	0.0649
15-19	531428	503091	105.6326	0.3839
20-24	318826	470153	67.8132	37.8193
25-29	445106	429185	103.7196	35.8963
30-34	423287	377645	112.086	8.3763
35-39	400660	325684	123.0211	10.9351
40-44	336097	265734	126.4796	3.4583
45-49	270331	209086	129.2918	2.8122
50-54	236988	186234	127.2528	2.0389
55-59	163501	143373	114.0389	13.2139
60-64	108910	105412	103.3184	10.7204
64-69	67351	74716	90.1426	13.1757
70-74	36587	48300	75.7494	14.3931
Total				153.8215
Average				S = 10.9872

	Male Age Ratio	Male Deviation from 100	Female Age Ratio	Female Deviation from 100
	103.5834	3.5834	103.8410	3.8410
	112.4238	12.4238	112.6139	12.6139
	91.6277	8.3722	79.2673	20.7326
	65.2974	34.7025	100.8613	0.8613
	119.9563	19.9563	101.2469	1.2469
	100.0955	0.0955	100.0557	0.0557
	105.5220	5.5220	106.2417	1.2417
	100.1798	0.1798	99.3825	0.6174
	94.3420	5.6579	92.3224	7.4775
	109.2533	9.2533	105.6769	5.6769
	94.5371	5.4628	98.3198	1.6801
	94.3548	5.6451	96.6667	3.3312
	92.5806	7.4192	97.0155	2.7844
	1283.7543	118.2747	1288.9146	62.1612
	98.7503	M = 9.0980	99.14267	F = 4.7816

From the table,

$$S = 10.9872$$

$$M = 9.0980$$

$$F = 4.7816$$

$$\therefore \text{UNAI} = 3(S) + M + F$$

$$= 3 \times 10.9872 + 9.0980 + 4.7816$$

$$= 46.8415$$

Comment: The value of UNAI is above 40. It indicates that the reported age-sex data for a given population is highly inaccurate. There is likely significant age misreporting. Common issues age heaping (clustering of ages ending in 0 or 5), digit preference etc. It suggests that respondents may not have reported their exact ages correctly, or enumeration recorded them inaccurately.

The dataset should be used with caution. Demographic analysis (like fertility, mortality etc.) may be biased. Data adjustment or smoothing techniques must be needed before serious use.

Problem: Using age distribution data for a West African country (1960), determine the extent of heaping on

- Digit '0' and '5'
- Digit '0' only
- Digit '5' only, using Whipple's index and interpret the result.

Age	Population	Age	Population
23	38687	45	44654
24	51239	46-49	77102
25	77141	50	52024
26-29	201450	51-54	44751
30	110379	55	14024
31-34	132196	56-59	45283
35	64091	60	38377
36-39	134140	61	4779
40	81515	62	7866
41-44	84422	Total	1304110

Solution: Whipple's index gives the relative preference for digit '0' and '5' while reporting age in the interval 23 to 62 years. It varies between 100 to 500.

If the heaping on terminal digit "0" and "5" is measured

$$\text{WI or Index} = \frac{\sum (P_{25} + P_{30} + \dots + P_{55} + P_{60})}{\frac{1}{5} \sum (P_{23} + P_{24} + \dots + P_{61} + P_{62})} \times 100$$

where, P_x = the no. of persons reporting their age as x years

If the heaping on terminal digit "0" is measured

$$WI = \frac{P_{30} + P_{40} + P_{50} + P_{60}}{\frac{1}{10} \sum (P_{23} + P_{24} + \dots + P_{61} + P_{62})} \times 100$$

If the heaping on terminal digit "5" only is measured

$$WI = \frac{P_{35} + P_{45} + P_{55} + P_{65}}{\frac{1}{10} \sum (P_{23} + P_{24} + \dots + P_{61} + P_{62})} \times 100$$

The index can be grouped in the following categories:

* If $WI = 100$, data has no heaping

* If $WI = 500$, data has only '0' and '5' digit heaping

* If $WI < 105$, data is very accurate

* If $WI = 105 - 110$, data is relatively accurate

* If $WI = 110 - 125$, data is good

* If $WI = 125 - 175$, data is bad

* If $WI > 175$, data is very bad

The relevant calculation given in the following table by using Ms Excel.

Age	Population	Age	Population
25	77141	25	77141
30	110379	35	64091
35	64091	45	44654
40	81515	55	14024
45	44654	Total	199910
50	52024	30	110379
55	14024	40	81515
60	38377	50	52024
		60	38377
Total	482205	Total	282295

Given that, total population = $\Sigma(P_{25} + P_{30} + \dots + P_{55} + P_{60})$
 $= 1304110$

From the table,

$$\Sigma(P_{25} + P_{30} + \dots + P_{55} + P_{60}) = 482205$$

$$P_{25} + P_{35} + P_{45} + P_{55} = 199910$$

$$P_{30} + P_{40} + P_{50} + P_{60} = 282295$$

(a) Age heaping on '0' and 5,

$$WI(0,5) = \frac{482205}{\frac{1}{5} \times 1304110} \times 100$$

$$= 184.8790$$

(b) Age heaping on digit '0' only,

$$WI(0) = \frac{282295}{\frac{1}{10} \times 1304110} \times 100$$

$$= 216.4656$$

(c) Age heaping on digit '5' only,

$$WI(5) = \frac{199910}{\frac{1}{10} \times 1304110} \times 100$$

$$= 153.2923$$

Comment:

Age Heaping	WI	Comment
Digit '0' and '5'	184.8790	very bad
Digit '0' only	216.4456	very bad
Digit '5' only	153.2923	bad

For question (a), it indicates very strong age heaping on ages ending in 0 and 5. For (b), respondents show a very strong preference for ages ending in 0. Whipple's index for digit '5' only is 153.2923, this also noticeable heaping, though less severe than for digit 0.

The age data suffer from substantial digit preference especially for ages ending in 0. Respondents were much more likely to report rounded ages (particularly multiples of 10), indicating very poor age reporting accuracy for that West African country. The heaping on digit 0 is stronger than on digit 5.

Problem: Use the Myer's blended index to assess the quality of age data given below :-

Digit	10-19	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
0	386	417	307	190	89	80	50	40	10
1	133	93	90	89	41	35	33	10	2
2	341	227	100	40	30	25	15	6	4
3	223	160	90	46	28	22	8	5	1
4	201	138	50	38	25	10	8	5	3
5	298	238	201	154	145	55	25	2	3
6	198	105	70	49	31	18	12	6	4
7	166	86	50	45	25	9	5	4	2
8	255	100	90	40	28	10	5	5	2
9	132	85	40	20	8	6	2	2	2

Solution: Myer's blended index procedure for computations:

1. Sum all the populations ending in each digit over the whole range i.e. 10-99.

2. Sum figures between ages 20-99.

3. Multiply the sums in (1) by coefficients; 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10.

4. Multiply the sums in (2) by coefficients from 9 descending to 0 i.e. 9, 8, 7, 6, 5, 4, 3, 2, 1, 0.

5. Add the product of (3) and (4), to obtain the blended sum.

6. Add up the blended sum in (5).

7. Find the percent (%) of the total blended sum of different digit ends.

8. Take the deviations of each % in (7) from 10. This result indicates the extent of concentration or avoidance of a particular digit.

						Blended	% :	Absolute
Sum1	Weight1	Product1	Sum2	Weight2	Product2	sum	Blended	Deviation
						(P ₁ +P ₂)	Population	from 10
1569	1	1569	1183	9	10647	12216	21.6945	11.6945
526	2	1052	393	8	3144	4196	7.4517	2.5482
788	3	2364	447	7	3129	5493	9.7551	0.2448
583	4	2332	311	6	2160	4492	7.9774	2.0225
478	5	2390	277	5	1385	3775	6.7040	3.2959
1134	6	6804	536	11	3344	10148	18.0219	8.0219
493	7	3451	275	3	885	4336	7.7003	2.2996
392	8	3136	226	2	452	3588	6.3719	3.6280
1535	9	4815	200	1	280	5095	9.0482	0.9517
297	10	2970	145	0	0	2970	5.2744	4.7255
Total						56309	100	39.4331

Interpretation of Myer's blended index:

MI	Interpretation	Quality of Age Data
0	No age heaping	Perfect quality
0-5	very little age preference	Very good
5-10	slight age heaping	Good
10-20	Moderate age heaping	acceptable
20-30	High age heaping	Poor
Above 30	very severe age heaping	Very Poor

$$\text{Myer's Blended Index} = \frac{1}{2} \sum |P_i - 10|$$

From table, $\sum |P_i - 10| = 39.4331$,
where, $P_i = \% \text{ blended population}$

$$\therefore \text{Myer's Blended Index} = \frac{1}{2} \times 39.4331 = 19.7165$$

Comment: The age data shows considerable age heaping. Digits 0 and 5 are strongly preferred, especially digit 0 with 21.73% and digit 5 with 18.04%, instead of the expected 10% each.

Our calculated Myer's index is 19.77, indicates moderate to high age heaping. The age data is fair or acceptable quality but not fully reliable age data. Strong preference for ages ending in 0 and 5.

Problem: The following table gives the census population of Bangladesh for the years 2011 and 2021.

	Age Group	Population	
		2011	2021
	0-4	195300	200000
	5-9	180700	185000
	10-14	165400	170090
	15-19	155200	165000
	20-24	148400	150400
	25-29	143300	145000
	30-34	140100	140600
	35-39	137200	135800
	40-44	130200	130800
	45-49	123700	125000
	50-54	114700	115600
	55-59	110800	113400
	60-64	93750	96500
	65-69	84782	85700
	70-74	73856	75300
	75-79	56500	60000
	80+	44000	45500

(a) Draw a population pyramid

(b) Compute intercensal growth rate using different growth functions.

(c) Find the time to double the population size of 2011.

(d) Calculate the projected population for the years 2041 and 2050.

(e) Make an overall comment

(a)

Solution: For drawing population pyramid the relevant calculation given in the following table:

Age Group	Population		% Population	
	2011	2021	2011	2021
0-4	195300	200000	-9.3093	9.3471
5-9	180700	185000	-8.6134	8.6461
10-14	165400	170090	-7.8841	7.9492
15-19	155200	165000	-7.3979	7.7113
20-24	148400	150400	-7.0737	7.0290
25-29	143300	145200	-6.8306	6.7766
30-34	140100	140500	-6.6781	6.5710
35-39	137200	135700	-6.5399	6.3467
40-44	130200	130300	-6.2062	6.1130
45-49	123700	125000	-5.8964	5.8419
50-54	114700	115600	-5.4674	5.4026
55-59	110800	113400	-5.2815	5.2998
60-64	93750	96500	-4.4687	4.5099
65-69	84782	85700	-4.0413	4.0052
70-74	73856	75300	-3.5204	3.5192
75-79	56500	60000	-2.6931	2.8041
80+	44000	45500	-2.0973	2.1264
Total	2097888	2139690	-100	100

graph:

Comment: The graph seems to represent an expansive population pyramid. It has a wide base—meaning there is a large proportion of children and young people. The pyramid gradually narrows toward the top—showing that the number of older people is smaller. This shape usually indicates high birth rates, rapid population growth and lower life expectancy compared to developed populations. These types of pyramids are seen in developing countries.

(b) For linear growth function,

$$r = \frac{P_t - P_0}{P_0 t}$$

where, P_t = Population at the end year

P_0 = Population at the current/last year

t = time period

r = growth rate

From the table, $P_t = 2139690$

$P_0 = 2097888$

$t = 10$ years

$$\begin{aligned} \therefore r &= \frac{2139690 - 2097888}{2097888 \times 10} \\ &= 0.00199 \end{aligned}$$

For geometric growth function,

$$\begin{aligned} r &= \left(\frac{P_t}{P_0} \right)^{\frac{1}{t}} - 1 \\ &= \left(\frac{2139690}{2097888} \right)^{\frac{1}{10}} - 1 \\ &= 0.00197 \end{aligned}$$

For exponential growth function,

$$\begin{aligned} r &= \frac{1}{t} \ln\left(\frac{P_t}{P_0}\right) \\ &= \frac{1}{10} \ln\left(\frac{2139694}{2097888}\right) \\ &= 0.00197 \end{aligned}$$

(c) Calculating the time to double the population size of 2011

For linear growth function,

$$t = \frac{1}{r} = \frac{1}{0.00197} = 507.6190 \text{ Years}$$

For geometric growth function,

$$t = \frac{\ln 2}{\ln(1+r)} = \frac{\ln 2}{\ln(1+0.00197)} = 351.3193 \text{ Years}$$

For exponential growth function,

$$t = \frac{\ln 2}{r} = \frac{\ln 2}{0.00197} = 351.3193 \text{ Years}$$

(d) Calculating the projected population for the years 2041 and 2050;

For linear growth function,

$$P_{30} = P_0(1+rt) = 2097888(1+0.00197 \times 30) = 2223294$$

$$P_{39} = 2097888(1+0.00197 \times 39) = 2260916$$

For geometric growth function,

$$P_{30} = P_0 (1+r)^t = 2097888 (1+0.00197)^{30} = 222809$$

$$P_{39} = 2097888 (1+0.00197)^{39} = 2265686$$

For exponential growth function,

$$P_{30} = P_0 e^{rt} = 2097888 e^{0.00197 \times 30} = 2225809$$

$$P_{39} = 2097888 \times e^{0.00197 \times 39} = 2265686$$

(e) Overall comments: At first we draw population pyramid. It seems like error in population pyramid. After that we calculate interannual growth rate for linear, geometric and exponential growth functions. The values are 0.199%, 0.197% and 0.197% respectively. In question no. (c), we compute the time to double the population size of 2011 for the above three cases, 501.8630 years, 351.3193 years and 351.3193 years are the values of that growth functions respectively. In this question, the values for geometric and exponential growth function are same. But for linear growth functions, it is relatively large. It seems inaccurate result. So, linear growth functions gives inaccurate result for predicting time. In next question, we calculate the projected population for the years 2041 and 2050 respectively for each growth function.

Problem: From the following table calculate GFR, TFR, ASFR_i, ASFR_f, GRR and NRR. Also given sex ratio at birth = 1.08 or male:female ratio = 52:48

Age	No. of Women	Total Birth	Survival Factor
15-19	4704852	148505	0.98
20-24	5037627	855994	0.977
25-29	4962900	1027579	0.974
30-34	3321030	501110	0.97
35-39	2798413	222705	0.966
40-44	2224489	61040	0.959
45-49	1678304	4934	0.948

Solution:

General Fertility Rate: GFR is defined as

$$GFR = \frac{\text{No. of live births that occur during a year}}{\text{Mid-year population of women aged 15-49}} \times 1000$$

In algebraic notations,

$$GFR = \frac{B}{W_{15-49}} \times 100$$

where,

B = No. of live births that occur during a year

W₁₅₋₄₉ = Mid-year population of women aged 15-49

Age-Specific Fertility Rate: ASFR_i is expressed symbolically,

$$ASFR_i = f_i = \frac{B_i}{W_i}$$

where, B_i = no. of births to women of i th age group

W_i = no. of women in the i th age group

f_i = Age-specific fertility rate of women in the i th age group

Total Fertility Rate: TFR for the population is given by

$$TFR = ACFR_{45-49} = 5 \sum_{i=15-19}^{45-49} f_i = 5 \sum_{i=15-19}^{45-49} ASFR_i$$

Age-specific Fertility Rate for Female Birth Only: It is defined as

$$ASFR_{f_i} = \frac{B_{f_i}}{W_i}$$

where, B_{f_i} = no. of female live births to mothers of i th age interval.

Gross Reproduction Rate: The formula for computing GRR is as follows:

$$GRR = 5 \sum_{i=15-19}^{45-49} \frac{B_{f_i}}{W_i} = 5 \sum_{i=1}^7 \frac{B_{f_i}}{W_i}$$

$$\text{OR, } GRR = TFR \left(\frac{B_f}{B_m + B_f} \right) = TFR \left(\frac{1}{1 + \text{Sex ratio at birth}} \right)$$

Net Reproduction Rate: The formula for calculating NRR:

$$NRR = \sum_{i=1}^7 f_i^d \left(\frac{5L_x}{L_0} \right)$$

where, f_i^d = Age-specific fertility rates based on only female births in the numerator

$\frac{5L_x}{L_0}$ = Survival rate base on the female life table

Interpretation of NRR:

1. $NRR=1$, means exact replacement or one daughter per woman.
2. $NRR<1$, denotes below-replacement fertility, where there are fewer daughters to succeed their mother's generation.
3. $NRR>1$, indicates above-replacement fertility - the future generation of potential mothers will be larger than the one that produced them.

The relevant calculation is given in the following table:

Age	No. of Women	Total Birth	Survival Factor	$ASFR_i$	Female Birth	$ASFR_{fi}$
15-19	4704852	148505	0.98	0.0315	71397	0.0151
20-24	5037627	855994	0.977	0.1699	411536	0.0816
25-29	4962900	1027579	0.974	0.2070	494028	0.0995
30-34	3321030	501110	0.97	0.1508	240918	0.0725
35-39	2798413	222705	0.966	0.0795	107070	0.0382
40-44	2224489	61040	0.959	0.0274	29346	0.0131
45-49	1678304	4934	0.948	0.0029	2372	0.0014
Total	24727615	2821867	6.774	0.6693	1356667	0.3218

Female Birth: Let, female birth = F

then male birth = $1.08 F$ [$\because$ Sex ratio at birth = 1.08]

So, total births = $F + 1.08 F = 2.08 F$

$$\Rightarrow F = \frac{\text{Total births}}{2.08}$$

From the table, $B = 2821867$, $\sum_{i=1}^7 ASFR_{fi} = 0.3218$

$W_{15-49} = 24727615$,

$$\sum_{i=1}^7 ASFR_i = 0.6693, \quad \sum_{i=1}^7 f_i^d \left(\frac{SL_i}{T_i} \right) = 0.3129$$

Therefore, $GFR = \frac{2821867}{24727615} = 114.118$

$TFR = 5 \times 0.6693 = 3.3469$

$GRR = 5 \times 0.3218 = 1.6091$

$NRR = 5 \times 0.3129 = 1.5647$

Comment: The value of GFR is 114.118, indicates approximately 114 births per 1000 women of reproductive age (ages 15-49). The population has a comparatively high birth rate among women of childbearing age. In the table, we compute ASFR_i and ASFR_f. A TFR of 3.3469 means on average, a woman is expected to give birth to about 3 or 4 children during her reproductive lifetime.

Since we get the value of $GRR = 1.6091$ which is greater than 1. So it indicates an increasing population size. The value of NRR of the above problem is 1.5647, which is greater than 1, so the data indicates above-replacement fertility, the future generation of potential mothers will be larger than the one that produced them.

Problem: The following table gives the population of a country for the year 2015, with the estimated number of births and deaths :

Age	Male		Female		Birth	
	Population	Death	Population	Death	Male	Female
0-4	442532	18623	434980	17308	-	-
5-9	419042	1809	416736	1709	-	-
10-14	393543	948	384616	1638	-	-
15-19	308269	1233	314056	1329	3578	3043
20-24	257852	1289	269340	1481	7293	6690
25-29	230629	1776	236187	1671	6775	6261
30-34	204188	1633	203477	1665	4233	4187
35-39	182270	1588	176534	1289	2999	2685
40-44	162509	1967	145037	1233	593	725
45-49	128784	2138	122946	1352	129	128
50-54	102971	1905	96589	1188	-	-
55-59	80717	2478	78311	1605	-	-
60-64	58899	3099	58142	1980	-	-
65-69	37797	2428	39049	2468	-	-
70+	45099	5981	48866	7175	-	-
Total	3055101	48895	3024866	45091	25600	24119

1. Calculate Crude death rate (CDR) of the total population for males and females.
2. Calculate GFR, ASFR, TFR and GRR.
3. Comment on your findings.

Solution: The CDR is the number of deaths in a given year per 1000 mid-year population i.e.

$$CDR = \frac{D}{P} \times 1000 \quad [CDR = \text{Crude Death Rate}]$$

where, D = no. of death in an area during a specified time

P = Mid-year population of that area

ASFR	ASFR (Female)	
-	-	This formula can be used separately for men and women.
-	-	
-	-	
0.0220	0.0106	The formulae of GFR, ASFR, TFR and GR are stated in the previous problem.
0.0519	0.0248	
0.0556	0.0269	
0.0413	0.0205	
0.0321	0.0152	
0.0090	0.0049	From the table, $D_m = 48895$, $P_m = 305510$ $D_f = 45091$, $P_f = 302486$ $W_{15-49} = 1467577$ [Using excel] $B = 49719$ $\sum_{i=1}^7 ASFR = 0.2143$ $\sum_{i=1}^7 ASFR_f = 0.1042$
0.0020	0.0010	
-	-	
-	-	
-	-	
-	-	
-	-	
-	-	
-	-	
0.2143	0.1042	

$$(1) CDR (\text{male}) = \frac{D_m}{P_m} \times 1000 = \frac{48895}{305510} \times 1000 = 16.004$$

$$CDR (\text{female}) = \frac{D_f}{P_f} \times 1000 = \frac{45091}{302486} \times 1000 = 14.907$$

$$(2) GFR = \frac{B}{W_{15-49}} \times 1000 = \frac{49719}{1467577} \times 1000 = 33.8782$$

ASFR_i and ASFR_{fi} have calculated in the table.

$$TFR = 5 \sum_{i=1}^7 ASFR_i = 5 \times 0.2143 = 1.0716$$

$$GRR = 5 \sum_{i=1}^7 ASFR_{fi} = 5 \times 0.1042 = 0.5212$$

(3) Comment on our findings: At first, we calculate crude Death Rate (CDR) for male and female. The values are 16.004 and 14.967 respectively. About 16 male deaths per 1000 male population per year. On the other hand, about 15 female deaths per 1000 female population per year. Male mortality is slightly higher than female mortality.

Then we calculate GFR = 33.8782, indicates approximately 34 births per 1000 women of reproductive ages. After that we compute ASFR_i and ASFR_{fi} in the table. A TFR of 1.0716 means on average, a woman is expected to give birth to about 1 child during her reproductive lifetime. GRR = 0.5212, indicates on average, a woman is expected to give birth to about 1 daughters during her reproductive lifetime.

Problem: From the following information find the singulate mean age at marriage (SMAM) for both male and female.

i	Age Group	Male never married s_i	Female never married s_i
1	10-14	100	93.8
2	15-19	93	43.4
3	20-24	61.5	11.2
4	25-29	20.6	3.4
5	30-34	5.3	1.0
6	35-39	1.4	0.6
7	40-44	1.0	0.3
8	45-49	1.1	0.2
9	50-54	0.2	0.1

Solution: The formula of SMAM is presented as follows

$$SMAM = \frac{N + 5 \sum_{i=1}^7 s_i - (50 \times \bar{s})}{100 - \bar{s}}$$

where, $N = 1000$

s_i = Percentage never married in each 5-years age group

$\bar{s} = \frac{1}{2}(s_8 + s_9)$ = Percentage never married at age 50 or in the final age group, often estimated from ages 45-49 and 50-54.

SMAM = singulate Mean Age at marriage.

SMAM for Male: By using excel,

$$N = 1000$$

$$\sum_{i=1}^7 S_i = 326$$

$$\bar{S} = \frac{1}{2}(S_8 + S_9) = \frac{1}{2}(4 + 0.3) = 2.15$$

$$\begin{aligned}\therefore \text{SMAM} &= \frac{1000 + 5 \times 326 - (50 \times 2.15)}{100 - 2.15} \\ &= 25.7792\end{aligned}$$

SMAM for female:

$$N = 1000$$

$$\sum_{i=1}^7 S_i = 185$$

$$\bar{S} = 0.3$$

$$\begin{aligned}\text{SMAM} &= \frac{1000 + 5 \times 185 - (50 \times 0.3)}{100 - 0.3} \\ &= 19.1574\end{aligned}$$

Comment: Men marry for the first time at about 25.8 years on average. On the other hand, Women marry for the first time at about 19.2 years on average.

Men tend to marry considerably later than Women in the given population studied. Women are entering marriage at a relatively young age compared to men.

Problem: For Bangladesh with the following information estimate growth rate of population:

$$\text{Population 1991} = P_1 = 89.9$$

$$\text{Population 2001} = P_2 = 111.45$$

Hence find out the doubling time.

Solution: For linear growth function,

$$r = \frac{P_2 - P_1}{P_1 t}$$

where,

P_1 = population in 1991

P_2 = Population in 2001

t = time period

r = growth rate

$$\begin{aligned}\therefore r &= \frac{111.45 - 89.9}{10 \times 89.9} \\ &= 0.023971\end{aligned}$$

For geometric growth function,

$$\begin{aligned}r &= \left(\frac{P_2}{P_1} \right)^{1/t} - 1 \\ &= \left(\frac{111.45}{89.9} \right)^{1/10} - 1 = 0.02172\end{aligned}$$

For exponential growth function

$$r = \frac{1}{t} \ln \left(\frac{P_2}{P_1} \right)$$

$$r = \frac{1}{10} \ln \left(\frac{111.45}{89.9} \right)$$

$$= 0.021488$$

Calculating the time to double the population size:

For linear growth function,

$$t = \frac{1}{r} = \frac{1}{0.023971}$$

$$= 41.71694 \text{ years}$$

For geometric growth function,

$$t = \frac{\ln 2}{\ln(1+r)} = \frac{\ln 2}{\ln(1+0.02172)}$$

$$= 32.25769 \text{ years}$$

For exponential growth function,

$$t = \frac{\ln 2}{r} = \frac{\ln 2}{0.021488}$$

$$= 32.25769 \text{ years}$$

Comment: Firstly, we compute the growth rate for linear, geometric and exponential growth function. The values are 2.40%, 2.17% and 2.15% respectively. Secondly, we calculate the time to double the population size. The required time to double the population 41.72, 32.26 and 32.26 years for the above cases.

Problem: From the following data

Age of Women x to $x+5$	Gross annual ASFR $f(x)$	Population of women surviving from birth to mid-point of age interval $P(x)$
15-19	0.063	0.788
20-24	0.192	0.778
25-29	0.128	0.748
30-34	0.110	0.723
35-39	0.080	0.698
40-44	0.045	0.668
45-49	0.007	0.635

Calculate

- (i) Gross reproduction rate GRR
- (ii) Net reproduction rate NRR
- (iii) Mean age at fertility/mean length of generation
- (iv) Describe the shape of the age-specific fertility rate curve and make a comment.

Solution: The formulae of GRR and NRR are stated in Practical no. 05.

The mean length of a generation in demography is calculated as follows:

$$MLG = \frac{\sum_{i=1}^7 x_i f_i^d \left(\frac{5}{T_0} \right)}{NRR}$$

where, x_i = Mid-point of i th age group

f_i^d = Age-specific fertility rates based on female births in the numerator

$$\frac{5L_x}{l_0} = \text{Survival rate based on the female life table}$$

$$NRR = \text{Net Reproduction Rate} = \sum_{i=1}^5 f_i \left(\frac{5L_x}{l_0} \right)$$

The calculation is given in the following table

Age	Mid-value, x	ASFR, $f(x)$	$P(x)$	$f(x) \cdot P(x)$	$x \cdot f(x) \cdot P(x)$
15-19	17	0.063	0.788	0.0496	0.8439
20-24	22	0.092	0.778	0.1026	2.2593
25-29	27	0.128	0.748	0.0957	2.5850
30-34	32	0.11	0.723	0.0795	2.5449
35-39	37	0.08	0.698	0.0558	2.0660
40-44	42	0.045	0.668	0.0300	1.2625
45-49	47	0.007	0.635	0.0044	0.2089
Total		0.565	5.038	0.4179	11.7708

Therefore,

(i) $GRR = 5 \times 0.565 = 2.825$

(ii) $NRR = 5 \times 0.4179 = 2.0898$

(iii) $MLG = \frac{5 \times 11.7708}{5 \times 0.4179} = 28.1626$

Comment: The value of GRR is 2.825 which indicates on average, a woman is expected to give birth to about 3 daughter during her reproductive lifetime. On the other hand $NRR > 1$, indicates above-replacement fertility, the future generation of potential mothers will be larger than the one that produced them. $MLG = 28.16$, means that on average it takes 28.1626 years for one generation to replace another within a population. That is, a new generation is produced every 28.1626 years in that population.

(IV) ASFR curve:

Comment: From the graph, the pattern can be interpreted as follow:

The ASFR is relatively low among women aged 15-19 years (0.063), indicating lower fertility during the teenage years. Fertility rises sharply and reaches its peak in the 20-24 age group (0.132). This suggests that women in their early twenties have the highest fertility level. There is only a slight decline in the 25-29 age group (0.128), meaning fertility remains high during the late twenties. After age 30, the ASFR begins to decline steadily. The very low fertility rate in the 45-49 age group reflects the biological decline in reproductive capacity at older ages. The graph indicates that childbearing is concentrated mainly among women aged 20-34 aged years, with peak fertility occurring in the early twenties.

Problem: Construct a life table from the following data

Age (x)	Probability of dying
0	0.020
1	0.014
2	0.013
3	0.012
4	0.011
5-9	0.018
10-14	0.006
15-19	0.009
20-24	0.010
25-29	0.014
30-34	0.011
35-39	0.021
40-44	0.031
45-49	0.051
50-54	0.059
55-59	0.089
60-64	0.117
65-69	0.198
70-74	0.252
75+	1.000

Solution: The columns of life table are defined and denoted as follows:

x = the ages of persons, i.e., $x = 0, 1, 2, 3, \dots$

l_x = the no. of persons living at age x

d_x = the no. of persons dying between x and $x+1$.

Therefore,

$$d_x = l_x - l_{x+1} \Rightarrow l_{x+1} = l_x - d_x$$

q_x = the probability that a person aged x dies within 1 year

thus,
$$q_x = \frac{d_x}{l_x} = \frac{l_x - l_{x+1}}{l_x} = 1 - \frac{l_{x+1}}{l_x}$$

$$\text{or, } d_x = l_x \cdot q_x$$

p_x = the probability that a life aged x survive to age $x+1$

$$p_x = \frac{l_{x+1}}{l_x} = \frac{l_x - d_x}{l_x} = 1 - q_x$$

L_x = the no. of persons living between ages x and $x+1$.

If we assume that the deaths are uniformly distributed between age x and $x+1$, then

$$L_x = \frac{1}{2}(l_x + l_{x+1}) = \frac{1}{2}(l_x + l_x - d_x) = l_x - \frac{d_x}{2}$$

T_x = total no. of persons living at any, given time, who are of age x or more. Thus

$$T_x = L_x + L_{x+1} + \dots + L_{w-1}$$

where, L_{w-1} = the function corresponding to the last age in the table.

e_x^0 = the expectation of life after reaching age x .

$$e_x^0 = \frac{T_x}{l_x} = \frac{\sum L_x}{l_x} = \frac{\frac{1}{2}l_x + l_{x+1} + l_{x+2} + \dots}{l_x}$$

The life table is constructed as follows:

Let, the initial ^{no. of} persons at age 0, $l_0 = 100000$

x	q_x	p_x	l_x	d_x	L_x	T_x	e_x^0
0	0.080	0.920	100000	8000	96000	1499231	14.9923
1	0.014	0.986	92000	1288	91356	1403231	15.2525
2	0.013	0.987	90712	1179	90122	1311875	14.4619
3	0.012	0.988	89533	1074	88996	1221753	13.6458
4	0.011	0.989	88459	9373	87992	1132757	12.8055
5-9	0.018	0.982	87486	1575	86698	1044785	11.9424
10-14	0.006	0.994	85911	515	85653	958087	11.1521
15-19	0.009	0.991	85396	769	85011	872434	10.2164
20-24	0.010	0.990	84627	846	84203	787423	9.3046
25-29	0.014	0.986	83781	1173	83194	703220	8.3936
30-34	0.011	0.989	82608	909	82153	620026	7.5056
35-39	0.021	0.979	81699	1716	80841	537573	6.5836
40-44	0.031	0.969	79983	2479	78743	457032	5.7141
45-49	0.051	0.949	77504	3953	75527	378289	4.8809
50-54	0.059	0.941	73551	4340	71381	302762	4.1163
55-59	0.089	0.911	69211	6160	66131	231381	3.3431
60-64	0.117	0.883	63051	7377	59363	165250	2.6208
65-69	0.198	0.802	55674	11023	50163	105887	1.9018
70-74	0.252	0.748	44651	11252	39025	55724	1.2479
75+	1.000	0.000	33399	33399	16699	16699	0.4999

Comment: The life table shows the survival and mortality pattern of the population by age. At birth, the probability of dying q_x is 0.080 with 100000 (say) survivors, while life expectancy is highest at age 1 (15.25 years). Mortality remains low during ages 10-34 years but rises sharply after age 60, reaching 0.252 at ages 70-74. Overall, survival decreases and mortality increases with advancing age.

Problem: Calculate net migration by forward method for the following observations of Bangladesh for the census, 1981-1991.

Age	Male Population (1981)	SR	Male Population (1991)
0-4	7449	0.9413	8857
5-9	7183	0.9505	9065
10-14	6226	0.9489	6902
15-19	4129	0.9436	4546
20-24	3244	0.9399	4098
25-29	3241	0.9378	4324
30-34	2492	0.9339	3367
35-39	2358	0.9272	3269
40-44	3694	0.9167	2455
45-49	2862	0.9005	1938
50-54	2690	0.8764	1636
55-59	1621	0.8404	1089
60-64	1949	0.7885	1226
65-69	915	0.7151	632
70-74	949	0.6662	669
75-79	399	0.4688	264
80+	1002	0.2633	416

Solution: To calculate net migration using the forward method, we follow these steps:

1. Expected/Projected Population (1991): This is the population expected to be in 1991 based on the male population in 1981

and the survival ratio.

2. Net Migration: Net migration for each age group is calculated by subtracting the expected population in 1991 from the observed male population in 1991.

Formulae:

Expected population (1991) = Male population (1981) $\times$ SR

Net migration = Observed population (1991) - Expected population (1991)

Age	Male Population (1981)	SR	Male Population (1991)	Expected Population (1991)	Net Migration
0-4	7449	0.9413	8857	7011.7437	1845.2563
5-9	7183	0.9505	9065	6827.4415	2237.5585
10-14	6226	0.9489	6902	5907.8514	994.1486
15-19	4129	0.9436	4546	3896.1244	649.8756
20-24	3244	0.9399	4093	3049.0356	1043.9644
25-29	3241	0.9378	4324	3039.4098	1284.5902
30-34	2492	0.9339	3367	2327.2788	1039.7212
35-39	2358	0.9272	3269	2186.3376	1082.6624
40-44	3694	0.9167	2455	3386.2898	-931.2898
45-49	2862	0.9005	1938	2577.231	-639.231
50-54	2690	0.8764	1636	2357.516	-721.516
55-59	1621	0.8404	1069	1362.2884	-273.2884
60-64	1949	0.7885	1226	1536.7865	-310.7865
65-69	915	0.7151	632	654.3165	-22.3165
70-74	949	0.6062	669	575.2838	93.7162
75-79	399	0.4688	264	187.0512	76.9488
80+	1002	0.2633	416	263.8266	152.1734
Total					7602.1874

* Positive net migration indicates a higher observed population in 1991 than expected, suggesting net in-migration.

* Negative net migration indicates a lower observed population in 1991 than expected, suggesting net out-migration.

Comment: The total net migration for the 1981-1991 period in Bangladesh is 7602.18, indicating positive net migration. This means that over the decade, more people moved into the population than left it, suggesting an overall net in-migration.

Problem: From the following data

Place of enumeration	Place of birth				Total
	Chittagong	Dhaka	Khulna	Rajshahi	
Chittagong	18353975	134513	21809	11666	18531263
Dhaka	311326	20719900	112689	65915	21209828
Khulna	75786	15622	13719526	32701	13843633
Rajshahi	45477	179351	26150	16761126	17012103
Total	18796560	21049386	13880174	16871407	70597527

- (i) Calculate in-migration rate, out-migration rate, net migration rate and gross migration rate.
 (ii) Calculate effectiveness ratio for each division.

Solution:

In-Migration Rate (IMR): The no. of people who moved into a division divided by the total population of the division.

$$IMR = \frac{\text{Total In-migrants}}{\text{Total Population}} \times 1000$$

Out-Migration Rate (OMR): The no. of people who moved out of a division divided by the total population of the division.

$$OMR = \frac{\text{Total out-migrants}}{\text{Total population}} \times 1000$$

Net Migration Rate (NMR): The difference between in-migrants and out-migrants, divided by the total population of the division.

$$NMR = (\text{In-migrants} - \text{Out-migrants}) / \text{Total population} \times 1000$$

Gross Migration Rate (GMR): The sum of in-migrants and out-migrants, divided by the total population of the division.

$$GMR = (\text{In-migrants} + \text{Out-migrants}) / \text{Total population} \times 1000$$

Effectiveness Ratio: The effectiveness ratio evaluates the proportion of net migration to gross migration, giving insight into how balanced migration is

$$\text{Effectiveness Ratio} = \text{Net Migration} / \text{Gross Migration}$$

The relevant calculation is given in the following table by using excel:

Place of enumeration	Place of birth				Total
Chittagong	Chittagong	Dhaka	Khulna	Rajshahi	
Chittagong	18263975	134513	21806	11666	18531963
Dhaka	311924	20719900	112689	65915	21209828
Khulna	75784	15622	13719526	32701	1384363
Rajshahi	45477	179351	26150	16761125	17012103
Total	18796560	21049386	13880174	16871407	70597527
Net Migration	432585	329486	1600648	110282	1033001

Comment: Migration patterns show Dhaka and Rajshahi as net gainers, attracting more people than they lose, while Chittagong and Khulna are net senders, experiencing higher out-migration than in migration.

In migration	IMR	Out migration	OMR	NMR	GMR	ER
167988	9.0647	432585	23.3426	-14.2778	32.4074	44.0574
109928	23.0991	329486	15.5345	7.5645	38.6336	19.5800
124107	8.9649	160648	11.6044	-2.6395	20.5693	12.8324
250978	14.7529	110282	6.4825	8.2703	21.2354	38.7459
1033001		1033001				